Wireless Data Transmission by Using RF Modules

Digital Logic Design (CSE345) Section:02

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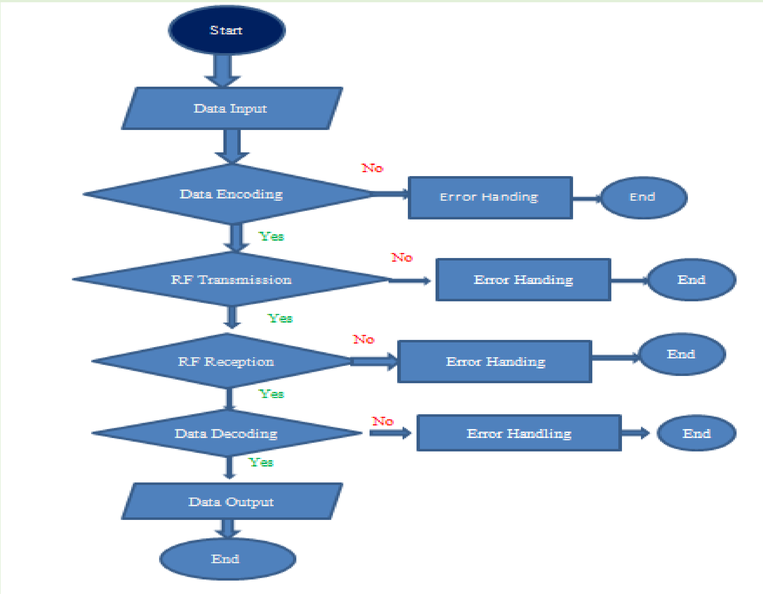
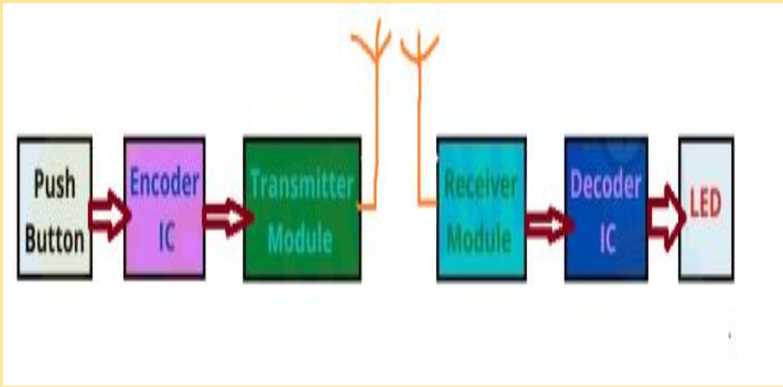
Introduction:

Wireless transmission is a form of unguided media. This data transmission process can be classified into two main types: radio frequency (RF) and optical. Similarly, we can assume that the heart of this wireless data transmission lies on RF (Radio Frequency) transmitter and receiver modules. In our academic project on wireless data transmission systems, we tried to built a wireless transmission system by using RF transmitter and receiver modules..

Methodology:

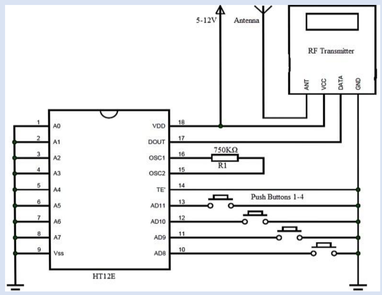


Flow diagram:

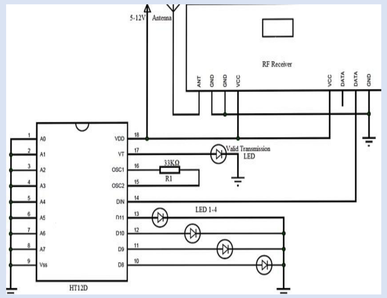


Internal Circuit Diagram:

RF Transmitter:



RF Receiver:



External Circuits of RF Transmitter and RF Receiver:

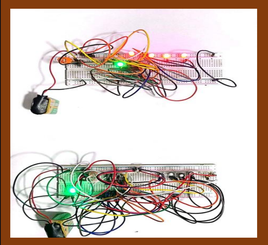


Fig-1

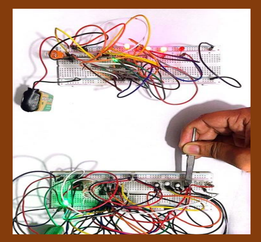


Fig-2

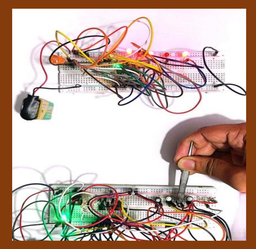


Fig-3

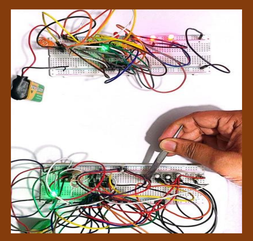


Fig-4

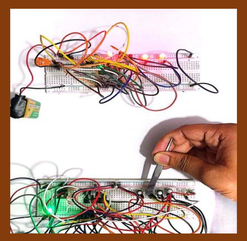
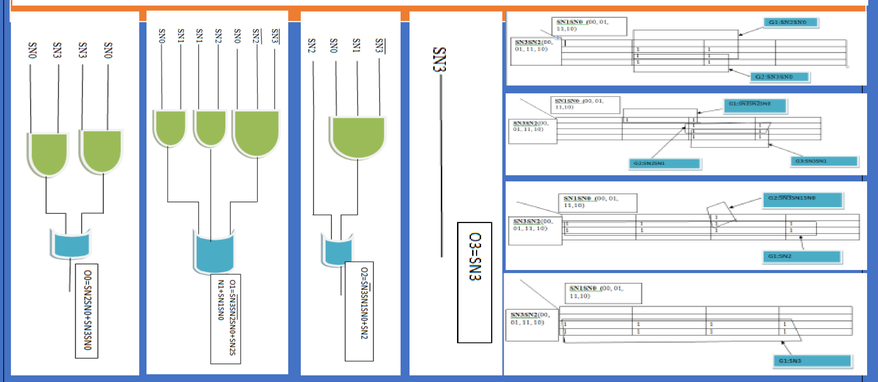


Fig-5

[Fig1:When active high on Transmitter circuit and all lights are on in the receiver circuit]
[Fig2,3,4,5:When active high is on one push button is off among 4 switch's also one LED is off .]

K-Map&Equations: From K-map we get equations for

$O_0 = SN_3SN_0 + SN_2SN_0$, $O_1 = SN_3'SN_2'SN_0 + SN_2SN_1 + SN_1SN_0$, $O_2 = SN_2 + SN_3'SN_1SN_0$, $O_3 = SN_3$



Truth table:

E	SN_3	SN_2	SN_1	SN_0	O_0	O_1	O_2	O_3
0	0	0	0	0	×	×	×	×
1	0	0	0	0	0	0	0	0
1	0	0	0	1	1	0	0	0
1	0	0	1	0	0	1	0	0
1	0	0	1	1	1	1	0	0
1	0	1	0	0	0	0	1	0
1`	0	1	0	1	1	0	1	0
1	0	1	1	0	0	1	1	0
1	0	1	1	1	1	1	1	0
1	1	0	0	0	0	0	0	1
1	1	0	0	1	1	0	0	1
1	1	0	1	0	0	1	0	1
1	1	0	1	1	1	1	0	1
1	1	1	0	0	0	0	1	1
1	1	1	0	1	1	0	1	1
1	1	1	1	0	0	1	1	1
1	1	1	1	1	1	1	1	1

Result:

When we provide Power supply to both circuits, we should notice that all LEDs will start glowing. Because the push-button pins (IC pin D8-D11) are **pulled up internally** by the Encoder IC. If we press one push-button the data pin is connected to the ground in the transmitter circuit, then the respective LED will be turned off in the receiver circuit.
For example, if we press **Switch 1 (S1)** which is connected to the **AD11 (pin13)** of the Encoder IC, then the **LED** will turn off which is connected to the **D11 (pin13)** of the Decoder IC.

Conclusion:

We successfully built this project by using RF transmitter and receiver. As we can see data is transmitted through the RF transmitter and then decoded that data on the RF receiver circuit and from the result of LEDs ON and OFF, we can say that our project is successful. The data transmission process is built to produce wireless transmission systems and in the near future, we will try to make this project more effective for our real-life methods.